

AI Agents That Actually Work

Building systems that survive contact with production

A 20-minute talk for developers

BG:dramatic-contrast

Most agent demos die in production

~80%

of polished demos never reach a real user

- Brittle prompts, no memory, flaky tools.
- Demo data is clean; production isn't.
- Latency and cost compound under load.

IMAGE:demo-vs-production-hero

Three pillars of agents that work

Planning, memory, and tool use — each non-optional.

SMARTART:three-pillars

Pillar 1 • Planning

Decompose before acting

- Turn a goal into a graph of sub-goals.
- Re-plan when the world changes — don't freeze the plan.
- Make intermediate state inspectable.
- Budget steps; fail fast when the plan stalls.



IMAGE: pillar-planning-
architecture

Pillar 2 • Memory

Short, long, and sideways

- Working memory: the live scratchpad.
- Episodic memory: what happened last time.
- Semantic memory: facts and embeddings.
- Forget deliberately — stale context is a bug.

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IMAGE: pillar-memory-diagram

Pillar 3 • Tool use

Typed interfaces beat prompt tricks

- Every tool has a schema — validate inputs and outputs.
- Treat tool errors as signal, not noise.
- Meter tool calls: rate-limit, cache, retry with backoff.
- Keep the tool catalogue small enough to reason about.

IMAGE: pillar-tool-use-sketch

Case study · A support-triage agent

Context

- Mid-sized SaaS, ~4,000 tickets/week.
- Previous rules engine misrouted ~22% of tickets.
- Agent has 4 tools: classify, lookup-customer, draft-reply, escalate.

22% → 6%

Misroute rate after rollout (8 weeks)

- Average reply latency: 3.1 min
- Escalation precision: 0.91
- Cost per ticket: \$0.04

How we rolled it out

Eight weeks from first prompt to production default.

SMARTART:rollout-timeline

Four things we got wrong first

Over-planned

Deep plans rotted as tickets changed mid-flight. Shorter horizons + re-plan won.

Under-memored

Agent forgot the same customer twice in one thread. Added a per-conversation store.

Tool sprawl

12 tools meant hallucinated calls. Cut to 4; precision jumped.

No kill switch

First week: one bad loop, \$400 in API spend. Now: step budget + hard cap.

Build small. Measure ruthlessly. Ship.

- Pick one workflow. Automate the narrow slice.
- Instrument every tool call. Review the traces weekly.
- Kill agents that don't earn their cost.

github.com/your-org/agents-that-work · @you

BG:calm-minimal